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CPCS-214 Computer Architecture and Organization

04 - Under Covers

Introduction

- We have looked below program to uncover underlying SW
- Let's open covers of computer to learn about underlying HW

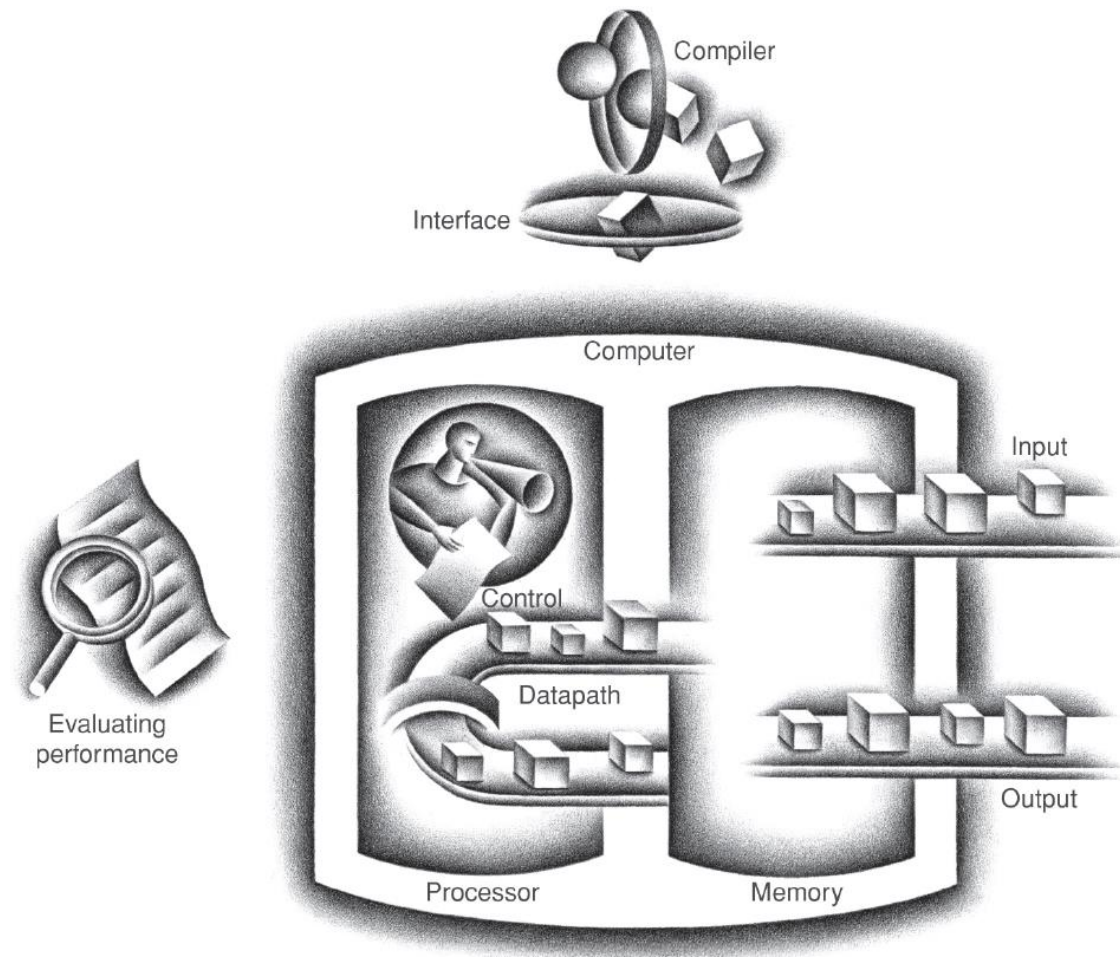
Basic Functions

- Underlying HW in any computer performs same basic functions
 - Inputting data
 - Outputting data
 - Processing data
 - Storing data

Computer Organization

- Five classic components
 - Input, output, memory, datapath, and control
 - Last two sometimes combined and called the processor
- This organization is independent of HW technology
- Can place every piece of every computer, past and present, into one of these five categories

Computer Organization



Through the Looking Glass

- Most fascinating I/O device is graphics display
- Most PMDs use liquid crystal displays (LCDs)
 - to get a thin, low-power display
- Most LCD displays use active matrix
 - Has tiny transistor switch at each pixel to precisely control current and make sharper images

Pixels

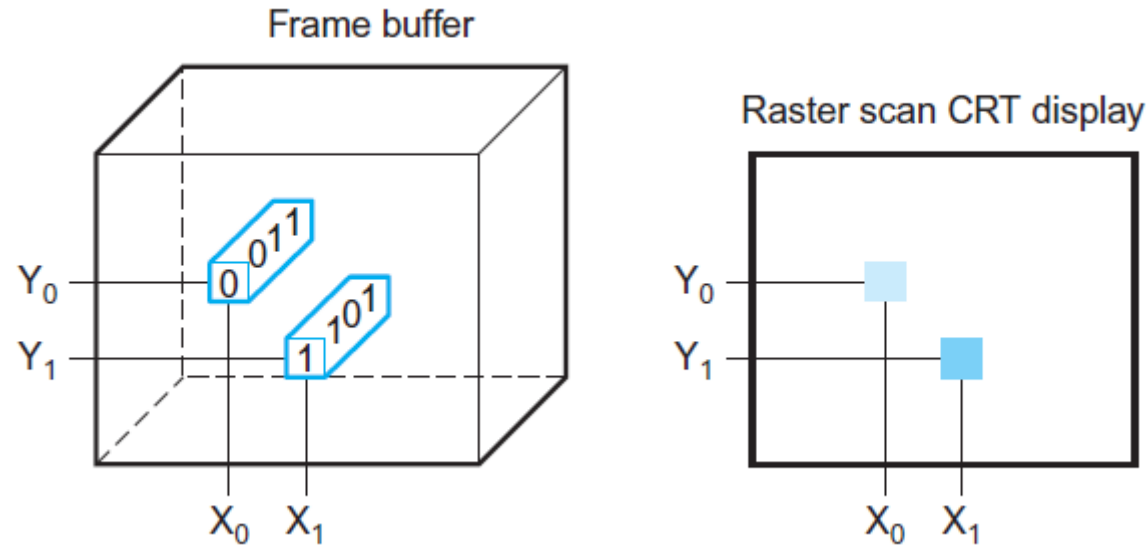
- Image is composed of a matrix of picture elements, **pixels**
- **Pixel**: smallest individual picture element
 - Screens are composed of hundreds of thousands to millions of pixels, which can be represented as a matrix of bits, called a *bit map*

Color Display

- Might use 8 bits for each of 3 colors
 - Red, Green, Blue (RGB)
 - 24 bits per pixel
- Permitting millions of different colors to be displayed
- RGB mask associated with each dot on display determines intensity of 3 color components in final image
- In color LCD, there are 3 transistor switches at each point

Frame Buffer

- Image to be represented on screen is stored in frame buffer
- Bit pattern per pixel is read out to graphics display at refresh rate

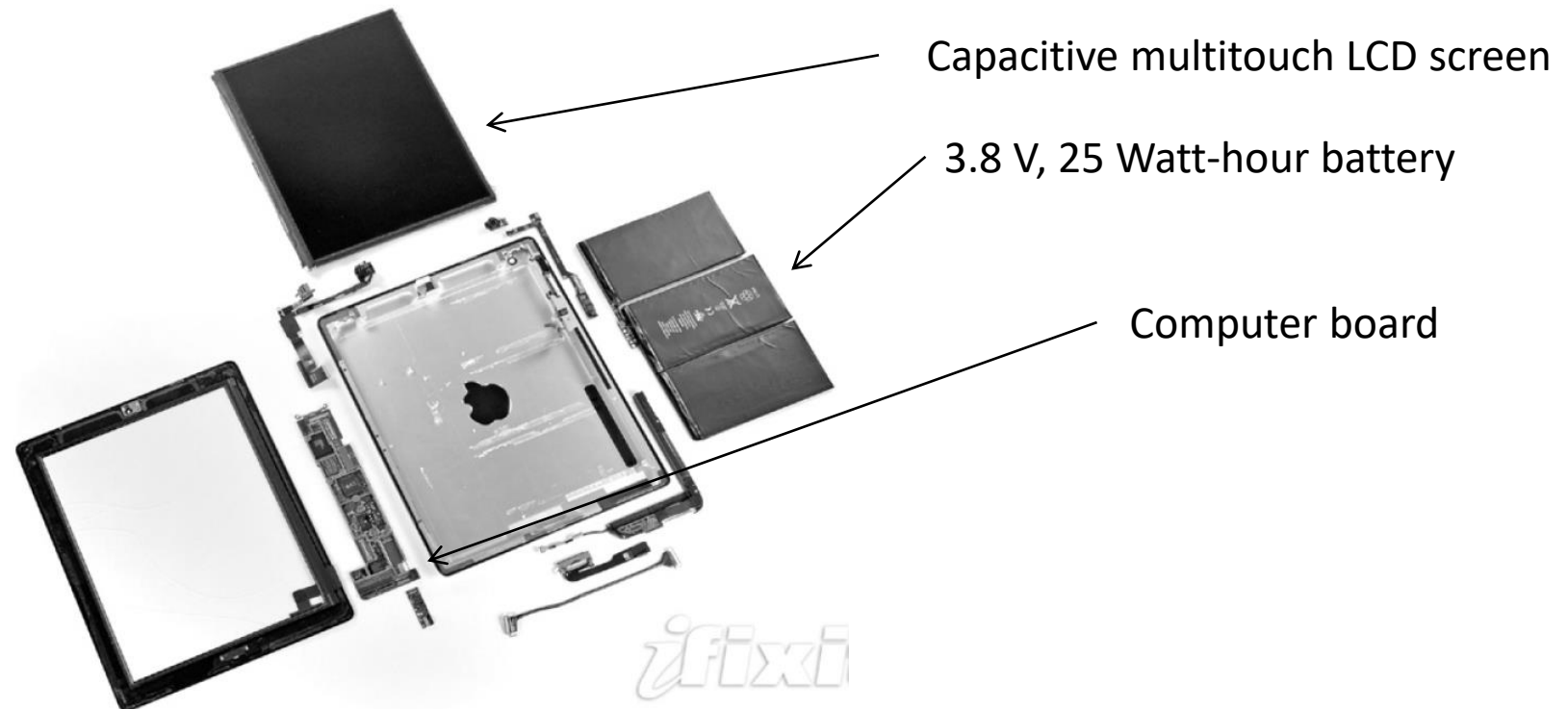


Touchscreen

- Tablets and smartphones of PostPC era have replaced keyboard and mouse with touch sensitive displays
- Advantage
 - Users pointing directly what they are interested in rather than indirectly with a mouse

Opening the Box

- Contents of Apple iPad 2 tablet computer



Opening the Box

- Of the five components of computer, I/O dominates this device
 - List of I/O devices includes
 - Multitouch LCD display, front camera, rear camera, microphone, headphone jack, speakers, accelerometer, gyroscope, Wi-Fi, and Bluetooth
- Datapath, control, and memory are tiny portion of components

Chips

- Small rectangles contain devices that drive our advancing technology
 - Called **integrated circuits (IC)** and nicknamed **chips**
- A5 package seen in middle contains two ARM processors that operate with a clock rate of 1 GHz



Processor

- Active part of computer, following instructions of a program to the letter
- Adds numbers, tests numbers, signals I/O devices to activate, and so on
- Occasionally, people call processor **CPU**
 - Central Processor Unit

Inside Processor

- Descending lower into HW, Figure reveals details of a μ P

Apple A5



Inside Processor

- Comprises two main components
 - Datapath performs arithmetic operations
 - Control tells datapath, memory, and I/O devices what to do according to the wishes of instructions of program

Memory

- A5 package includes two memory chips
 - Each with 2 gibibits of capacity, thereby supplying 512 MiB
- Where programs are kept when they are running
- Also contains data needed by the running programs
- Built from DRAM chips

DRAM

- Stands for **Dynamic Random Access Memory**
- Multiple DRAMs are used together
 - to contain instructions and data of a program
- *RAM* memory accesses take basically same amount of time no matter what portion of memory is read
 - In contrast to sequential access memories, such as magnetic tapes

Cache

- Inside processor
- Consists of a small, fast memory
- Acts as a buffer for DRAM
- Nontechnical definition of *cache* is safe place for hiding things
- Built using different memory technology
 - Static Random Access Memory (SRAM)
- Faster but less dense, and hence more expensive, than DRAM
- SRAM and DRAM are two layers of the memory hierarchy

Safe Place for Data

- If to lose power to computer, everything would be lost
 - Because memory inside computer is **volatile**
 - i.e., when it loses power, it forgets
- In contrast, DVD disk doesn't forget data when turn off power to DVD player
 - **Nonvolatile**

Safe Place for Data

- **Volatile** memory used to hold data and programs while they are running
- **Nonvolatile** memory used to store data and programs between runs
- To distinguish between the two
 - The term **main memory** or **primary memory** is used for the former, and **secondary memory** for the latter

Secondary Memory

- Forms the next lower layer of **memory hierarchy**
- DRAMs have dominated main memory since 1975
- Magnetic disks dominated secondary memory starting even earlier

Flash Memory

- Nonvolatile semiconductor memory
- Because of their size and form factor
 - PMDs use instead of disks
- **Slower**, but much **cheaper** than DRAM
- Although costing more per bit than disks, it is
 - smaller
 - comes in much smaller capacities
 - more rugged
 - more power efficient

Flash Memory

- Standard secondary memory for PMDs
- Alas, unlike disks and DRAM, flash memory bits wear out after 100,000 to 1,000,000 writes
 - File systems must keep track of number of writes and have a strategy to avoid wearing out storage, such as by moving popular data

Communicating with Other Computers

- Networks interconnect whole computers
- Allowing computer users to extend power of computing by including communication
- Backbone of current computer systems

Networks Advantages

- *Communication*
 - Information is exchanged between computers at high speeds
- *Resource sharing*
 - Rather than each computer having its own I/O devices, computers on the network can share I/O devices
- *Nonlocal access*
 - By connecting computers over long distances, users need not be near the computer they are using

Networks Performance

- Vary in length and performance
- Cost of communication increasing according to both
 - Speed of communication
 - Distance information travels

Ethernet/LAN

- Most popular type of network
 - It can be up to a km long and transfer at up to 40 gigabits per second
- Example of **local area network (LAN)**
- Its length and speed make Ethernet useful to connect computers on the same floor of a building
- LANs are interconnected with **switches** that can also provide routing services and security

WAN

- Cross continents
- Backbone of Internet, which supports web
- Typically based on optical fibers
- Leased from telecommunication companies

Networks Impact

- Have changed face of computing in the last 30 years
 - Becoming much more ubiquitous
 - Making dramatic increases in performance

Networking Technologies

- Networking technology improved: much cheaper and much higher capacity
 - First standardized LAN technology
 - Maximum capacity (bandwidth) of 10 million bps
 - Shared by tens of, if not a hundred, computers
 - Today, LAN technology offers
 - Capacity of from 1 to 40 giga bps
 - Shared by at most a few computers

Optical Communications Technology

- Allowed similar growth in capacity of WANs from
 - 100s of kilobits to gigabits
 - 100s of computers connected to a worldwide network to millions of computers connected

Wireless Technology

- Innovation reshaped way computers communicate
 - Widespread
 - Enabled PostPC Era
- Called by standard name: 802.11
- Allows for transmission rates from 1 to nearly 100million bps
- Quite a bit different from wire-based networks
 - All users in an immediate area share airwaves